Laboratory 8 – Exercises

1. Estimate the following linear model $y = Xw + b + \epsilon$, where $X \in \mathbb{R}^{2000\times 3}$ consists of 2000 examples, each consisting of 3 features sampled from a standard normal distribution and ϵ obeys a normal distribution with mean 0 and standard deviation of 0.1. The true weights of the model will be $w = [0.15, -0.2, 0.1]^T$ and $b = 0.2$. Create the linear regression model (**do not use PyTorch's predefined layers**) and train it as follows:

➢ set the weights and bias to 3;

➢ set the batch size to 20;

➢ **implement** the mean squared error loss;

➢ **implement** the mini-batch gradient descent optimizer, $lr = 0.02$ (do not use the torch.optim package);

➢ set the total number of epochs to 40.

After the model is trained, compare the learned model parameters with the actual parameters.

2. Implement the softmax regression model in order to classify the images from the Imagenette dataset (10 classes) as follows:

➢ download the Imagenette train images;

➢ split the train images as follows: validation-4456, training-4456, and the rest of the images represents the testing dataset;

➢ set the batch size to 100;

➢ resize the images to $(300, 100)$ resolution;

➢ initialize the weights at random with zero mean and standard deviation 0.01;

➢ use the cross-entropy loss function;

➢ use mini-batch gradient descent with a learning rate of 0.1 as the optimization algorithm;

➢ set the number of epochs to 5;

Evaluate the trained model on the test set.

3. Classify the Imagenette dataset (10 classes) using an MLP as follows:

➢ download the Imagenette train images;

➢ split the train images as follows: validation-5000, training-4000, and the rest of the images represents the testing dataset;

➢ set the batch size to 150;

➢ the MLP consists of five hidden layers of 128 units with the rectified linear unit activation function and a dropout layer after the third hidden layer (dropout probability 35%);

➢ set the weights from a normal distribution with $std = 0.01$ and the biases to 0;

➢ set learning rate to 0.15;

➢ train the model for 15 epochs.

Evaluate the trained model on the test set.